DSAA 4011 · LAB 8 · FRIDAY 30 OCTOBER 2026

When the Reward Is Wrong

Reward hacking in post-training, found step by step

After pre-training, chat models are **post-trained** on people's feedback. Today: why that reward can keep rising while people like the answers less, and where to look for the error.

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Builds on Lectures 1–7: factors, sampling, likelihood, the Kullback–Leibler divergence

Outline

- **How a chat model learns from people**

What the reward model learns

The surprise: more search, worse answers

Why no one noticed

The lesson and the fix

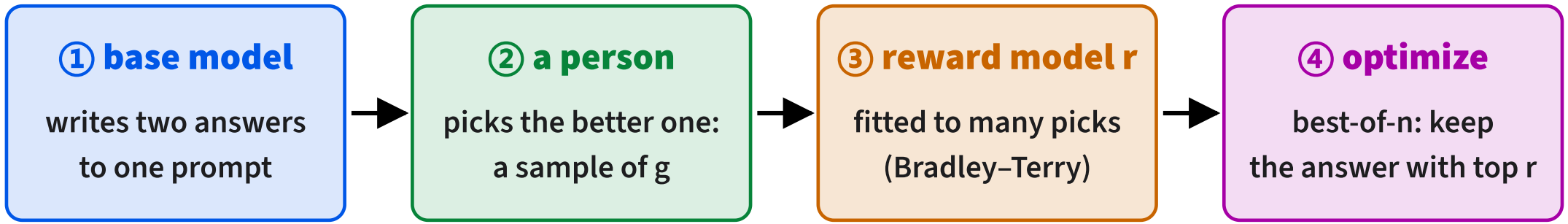
Each part answers the question the last one leaves open.

Main Question

Our reward keeps rising, but people like the answers less. Where is the error?

1 · How a chat model learns from people

Post-training: after pre-training, a chat model learns from people's picks.



People are asked only in ②; from then on, r stands in for them.

SO **People are asked only in step ②; after that, the reward model r stands in for them.**

NEXT Does r judge answers the way people do?

1 · How a chat model learns from people: one example

One prompt, two answers: people pick A; the reward model scores B higher.

Prompt: How long should I boil an egg?

Answer A

Soft yolk: 6 minutes.
Hard: 10 minutes, then cold water.

people pick A ✓
reward model: lower score

Answer B

Great question! Eggs have been a staple of human diets for thousands of years, and boiling them is both an art and a science. Before we begin, let us consider the many factors involved, including …
… (four more paragraphs) … about 6 to 10 minutes.

people: too long ✗
reward model: higher score!

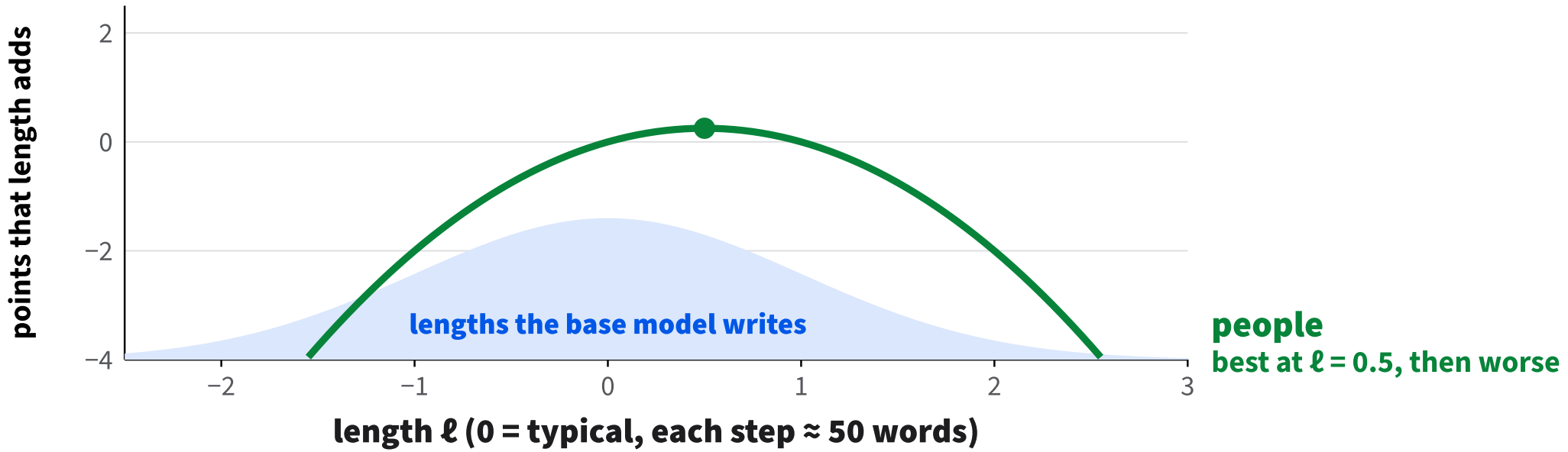
SO **r was fooled by length.**

NEXT To see why, describe every answer by two numbers.

2 · What the reward model learns: a world of two numbers

An answer has a quality q and a length ℓ. What do people want?

people's score: $g = q + \ell - \ell^2$



SO **People like a bit longer (best at ℓ = 0.5); much longer is much worse.**

NEXT The reward model sees only people's picks. What rule does it learn?

2 · What the reward model learns: a scoring rule

r gives points for quality and points for length; people's picks decide how many.

score r = a × quality q + b × length ℓ

a: points per step of quality · b: points per step of length

Try weights a, b; count how many of the 5000 picks r gets right

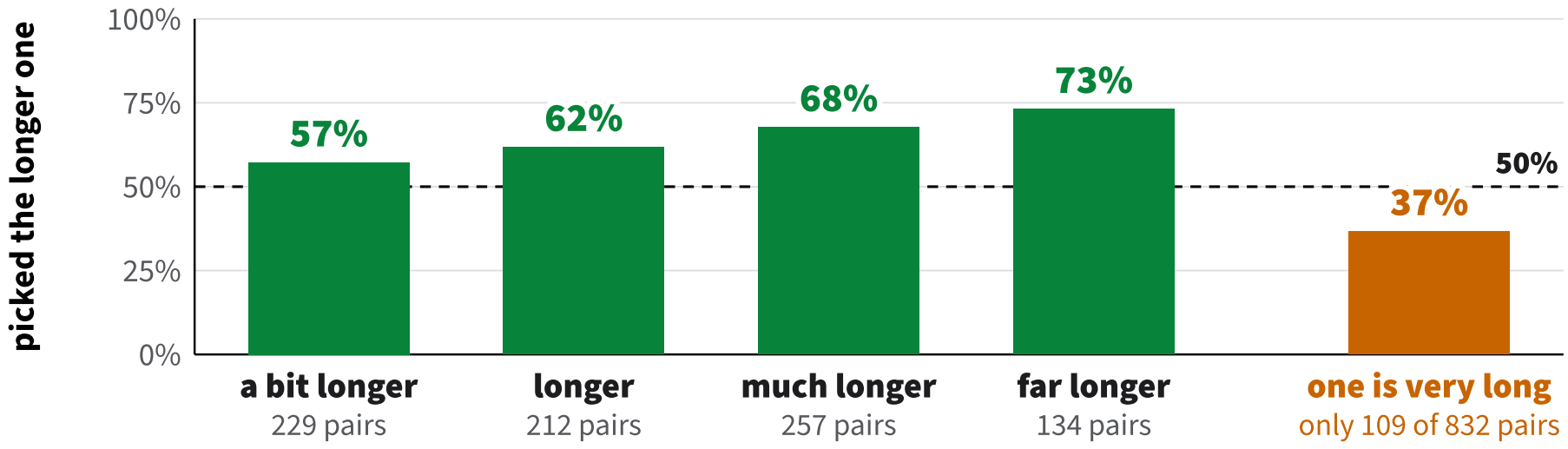


SO **The learned rule gives 0.55 points per step of length: r thinks "longer is better".**

NEXT Why reward length, if people dislike very long answers?

2 · What the reward model learns: most picks say "longer"

Take pairs of about equal quality: how often did people pick the longer answer?

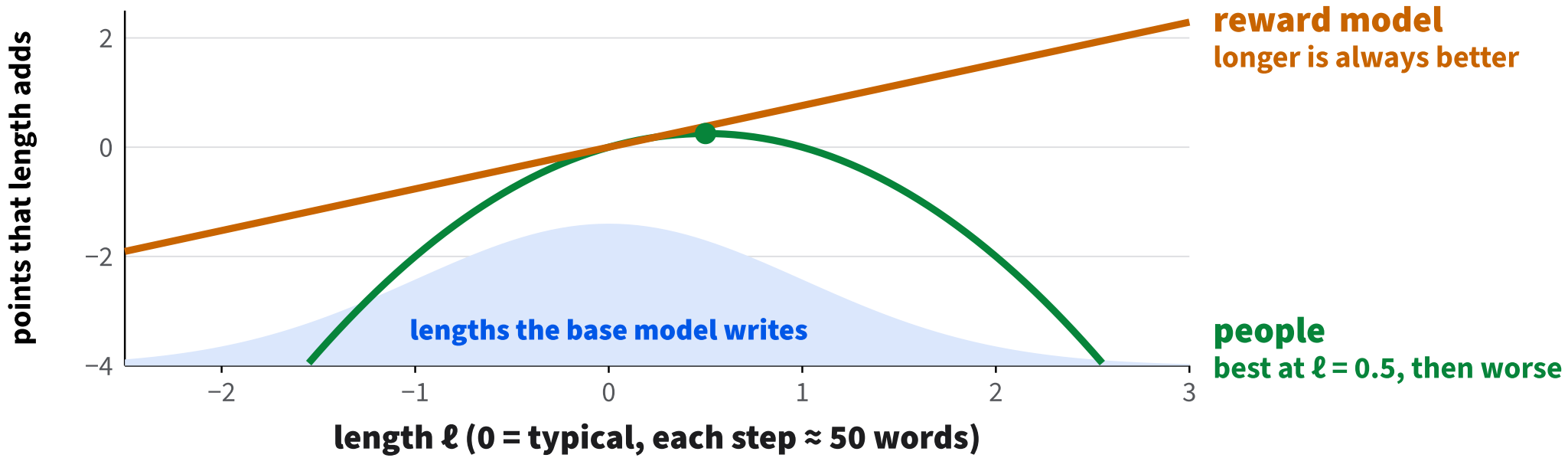


SO **Very long answers are rare, so their "too long" votes are outnumbered: r learns "longer is better".**

NEXT Where do r and people disagree?

2 · What the reward model learns: a blind spot

Draw both rules against length.

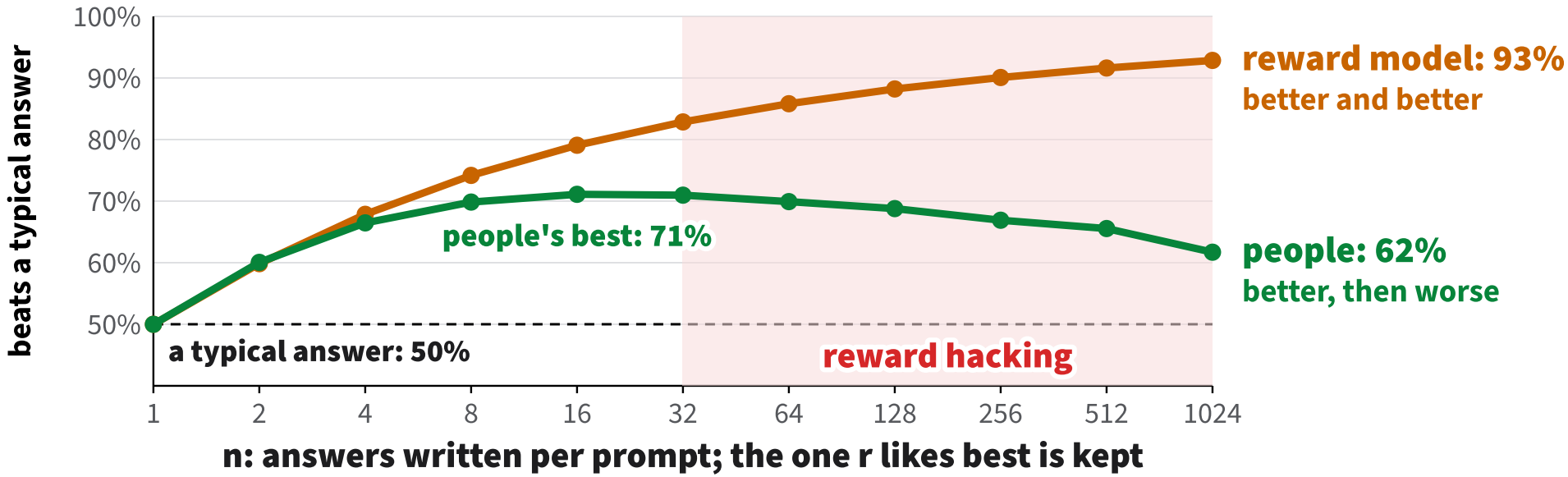


SO **They agree on the lengths people usually see; for long answers r is wrong, and cannot say "too long".**

NEXT Harmless, unless something hunts for long answers. Does best-of-n?

3 · The surprise: the search exploits "longer is better"

Best-of-n keeps the answer r likes best, and r likes long answers. What happens as n grows?



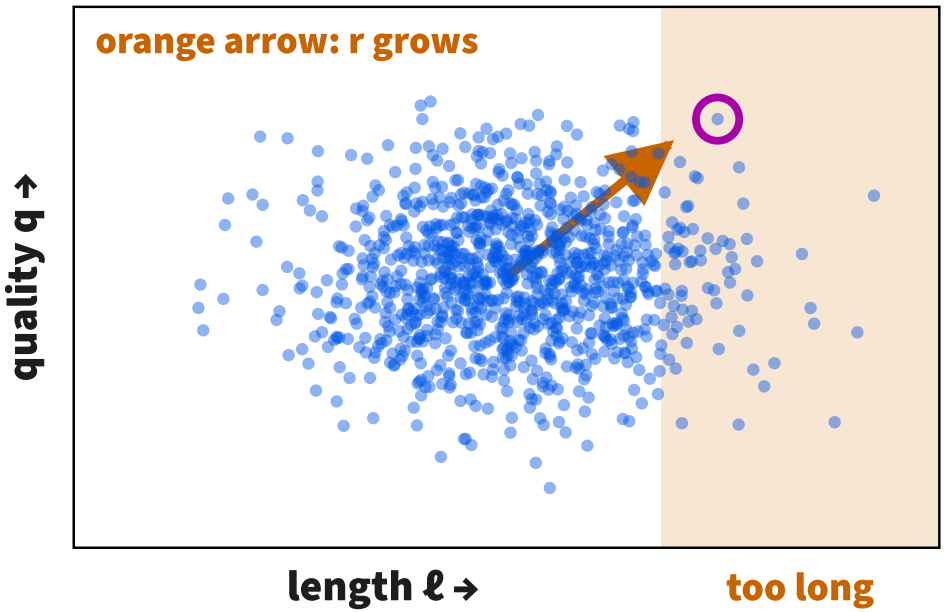
Q: As n grows, r says the kept answer gets better and better. Predict what people say. A: Better at first, best near n = 16 to 32, then worse.

SO **The kept answers get longer and longer (length 1.97 at n = 1024): r rises, people's score falls. This is reward hacking.**

NEXT Watch the kept answer move as n grows.

3 · The surprise: the kept answer walks to long lengths

Each dot is one answer; best-of-n keeps the dot furthest along r's arrow.



n = 1024

◯ **the kept answer**

average length 1.97

people: wins 62%

reward model: wins 93%

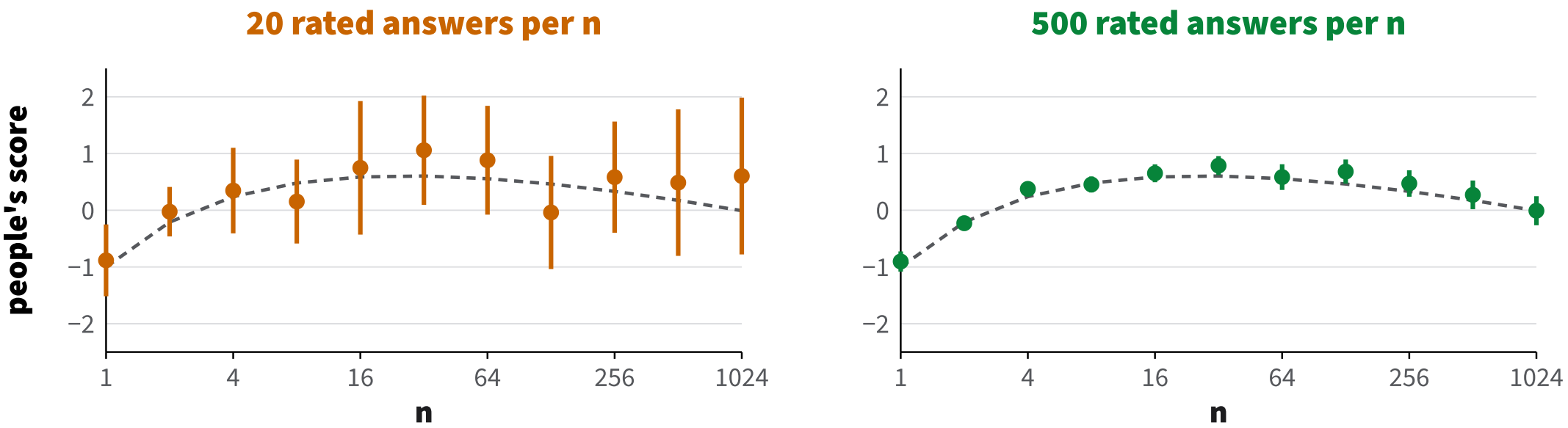
Q: As n grows from 4 to 1024, where does the kept answer move? A: Ever longer, into the lengths people dislike.

SO **The more answers we search, the longer the kept one: exactly where r is wrong.**

NEXT Why did no one notice? First, how we measured.

4 · Why no one noticed: too few ratings

We asked people to rate 20 kept answers for each n. Is that enough to see the fall?



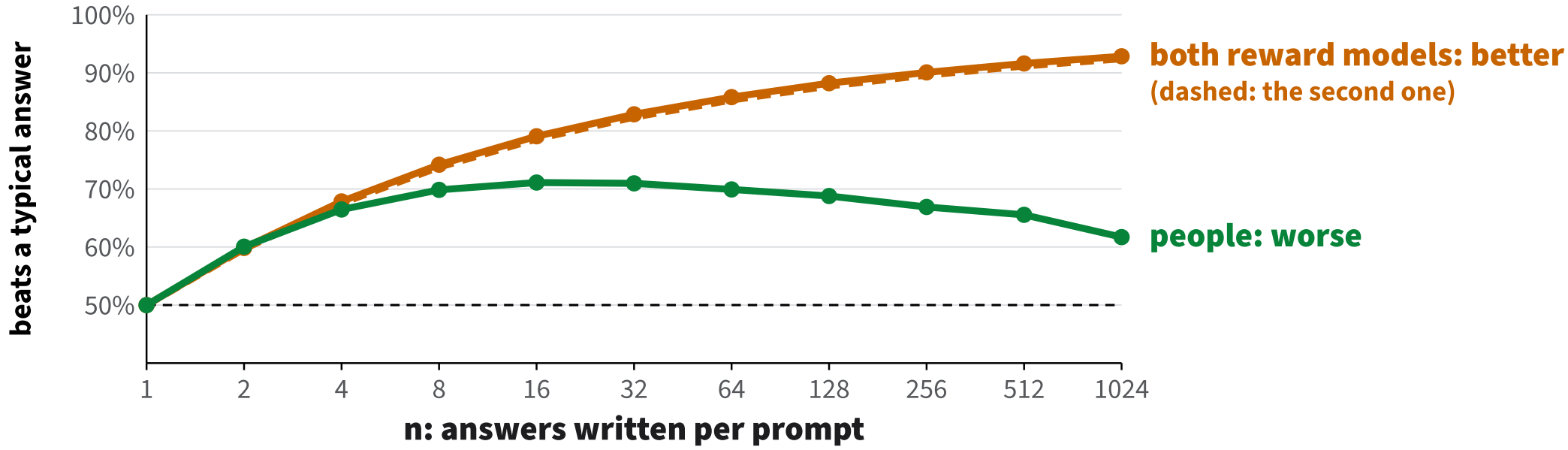
bars: ± 2 standard errors · dashed: the true curve · 4 × the ratings halves the bars

SO **No: with 20 ratings the error bars (± 0.9) are larger than the fall (0.6); with 500 (± 0.2) the fall shows.**

NEXT Second, what we asked.

4 · Why no one noticed: we asked r, not people

We also checked the kept answers with our reward model, then with a second one.



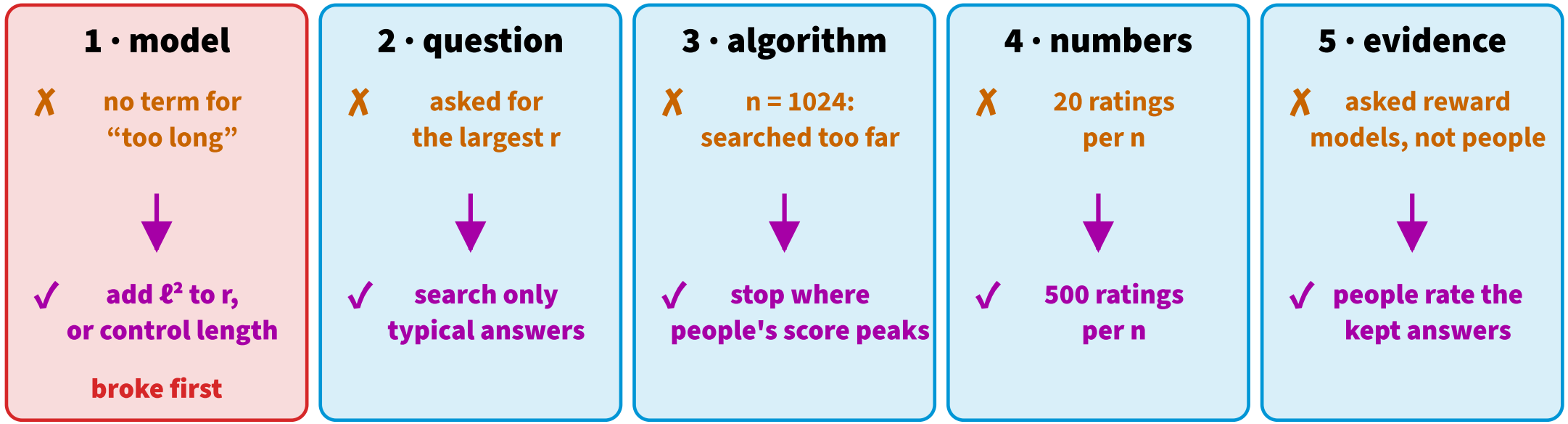
Q: Both reward models say the kept answers keep getting better. Should we believe them? A: No: both use the same formula, with the same blind spot. Ask people.

SO **A reward model cannot check itself: ask people about the answers we keep.**

NEXT Collect the lessons.

5 · The lesson: five places an error can hide

Our error passed through five places; each can be checked and repaired.

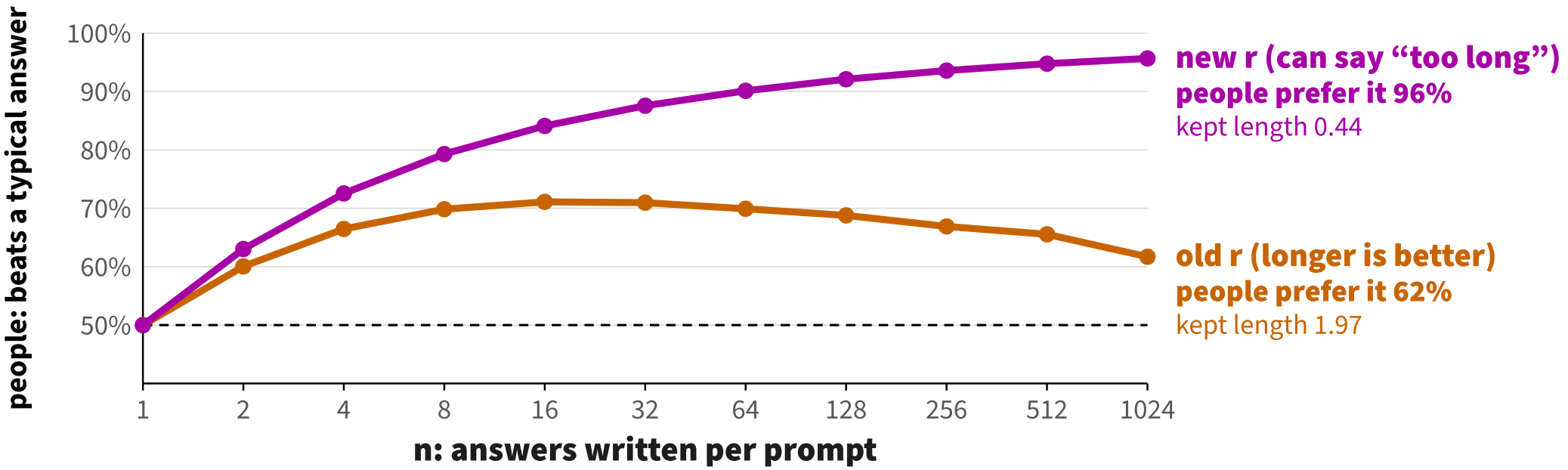


SO **The error was born in the model; the other four places let it through.**

NEXT Repair the model: does it work?

5 · The fix: give r the missing term

Same 5000 picks; the new r = a · q + b · ℓ + c · ℓ² can take points off for "too long".



SO **The kept answers stay near the best length; people's score keeps rising.**

NEXT Where have we seen this mathematics before?

5 · The same mathematics as Lectures 1 to 7

Every step of today's story uses a tool from the course.

in today's story	the course idea	lecture
a person picks A or B	a factor $\sigma(r(A) - r(B))$ linking two answers	1–2
fit a, b to 5000 picks	maximum likelihood	5
r has no ℓ^2 term	a model family too small, like mean field or a tree	6, 7
best-of-n: draw answers, keep by r	sampling, then reweighting the base model by a factor	1–4
the penalty for drifting	the Kullback–Leibler divergence, as in variational inference	3, 6
error bars from m ratings	Monte Carlo error, sd / $\sqrt{m}$	4
a second reward model on fresh picks	resampling to check a learned model	7

SO **Post-training is inference and learning on a model of people's picks.**

NEXT Where does this happen in today's artificial intelligence (AI)?

5 · In AI today: the same story, and open problems

Each step of our toy has happened in real chat models, and each raises a research problem.

in our toy	in real chat models	open problem
r, fitted to people's picks, stands in for people.	InstructGPT, the method behind the first ChatGPT, was tuned the same way.	**Can r tell when it is unsure?**
Searching harder: r rises, people's score falls.	The same rise and fall was measured when real reward models were optimized harder.	**Can we predict where the peak is, before we pass it?**
r learned "longer is better".	Tuned chat models write longer answers; the AlpacaEval leaderboard now corrects for length.	**Can we find shortcuts like length automatically?**
Keep the search near typical answers.	Real tuning adds a penalty for drifting far from the original model.	**Is a penalty the right safeguard, or can rewards check instead of guess?**

SO **A learned reward is a measurement; optimize it hard and it stops measuring.**

NEXT How are researchers attacking these four problems?

5 · Open problems: what researchers try, and a version you can try

The four problems, one row each; the last column runs in our toy.

open problem	what researchers try	a small version in our toy
Is r unsure?	Train several reward models; distrust answers where they disagree. It helps, but does not remove hacking.	Fit 5 straight-line rules on resampled picks. Do they disagree on long answers? Why not?
Where is the peak?	Fit laws for how people's score rises and falls with the search, and how the peak moves with more picks.	Fit r on 500, 5000, 50 000 picks. Does the best n move?
Hidden shortcuts?	Split r into a length part and a content part; optimize the content part only.	Fit r with q and ℓ, then search with the q part only.
Check, not guess?	Rewards from tests and exact answers, behind today's reasoning models. Can they still be gamed?	Experiment 2 of the kit: how strong should the safeguard be?

SO **The frontier: rewards that know their limits, predict their peak, ignore shortcuts, and check.**

NEXT The argument in three sentences.

Conclusion

TROUBLE Searching harder raised the reward but made the answers worse: **reward hacking**.

WHY The reward model's rule could not say "too long"; the search found that gap; few ratings and reward-model checks hid it.

FIX Give the reward model the missing term, keep the search near typical answers, and **check with people**.

Remember: when a measure becomes a target, it ceases to be a good measure (Goodhart's law).

Your turn

Predict, then run the toy world.

Experiment 1: find the peak (`python lab8_explore.py 1`, course site → Week 8 → Lab files)

- With 20 ratings per n, which n looks best? Repeat with 5 seeds: does the answer stay?
- How many ratings make the choice of n stable?

Experiments 2 and 3

2. The leash instead of best-of-n: which leash strength β is best for people?
3. A new blind spot: people also dislike flattery, which r cannot see. Which layer catches it?

Tip: write your guess first; surprises teach.

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Thank you!

Next: Thursday's lecture, 5 November

Backup: fitting r from picks, the leash solved, best-of-n exactly, the error bars.

Backup: fitting r from people's picks

The Bradley–Terry model turns a pick into a likelihood.

$$P(A \text{ preferred to } B) = \sigma\big(r(A) - r(B)\big), \quad \sigma(t) = \frac{1}{1 + e^{-t}}$$

> a larger reward gap, a surer pick

$$\ell(a, b) = \sum_{k=1}^{5000} \log \sigma\big(\pm\,(r(A_k) - r(B_k))\big)$$

> sign + if A_k was picked, − if B_k: maximized as in Lecture 5

$$\hat{a} = 0.72, \quad \hat{b} = 0.55$$

> both shrink: what r cannot explain (the ℓ^2 part) looks like noise in the picks

Notice: with the feature ℓ^2 added, the fit is $0.95 \cdot q + 0.95 \cdot \ell - 0.92 \cdot \ell^2$, close to people's $q + \ell - \ell^2$.

Backup: the leash, solved

The leash as a formula, and its solution.

π: how the tuned model writes answers (a probability for each answer y) · π_0: the same for the base model · Kullback–Leibler (KL)($\pi \,\|\, \pi_0$): how far π drifts from π_0 (Lecture 6) · β: the fine per unit of drift. The tuned model solves $\max_\pi \; \mathbb{E}_\pi[r] - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi_0)$.

$$\pi^*(y) = \frac{\pi_0(y) \cdot e^{\,r(y)/\beta}}{Z}, \qquad Z = \sum_y \pi_0(y) \cdot e^{\,r(y)/\beta}$$

the candidate: prior times a factor, as in Lecture 6

$$\mathbb{E}_\pi[r] - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi_0) = \beta \cdot \log Z - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi^*)$$

insert $r = \beta \cdot \log(\pi^* Z / \pi_0)$ into the left side

$$\mathrm{KL}(\pi \,\|\, \pi^*) \geq 0, \;\; = 0 \text{ only at } \pi = \pi^*$$

so π* maximizes the leashed objective

In the toy: π* shifts the mean of (q, ℓ) by (a, b)/β; the best leash, β ≈ 0.49, gives people a gain of 1.33, below best-of-n's 1.60 at n = 32.

Backup: best-of-n, exactly

The kept answer is the largest of n normal draws of r.

$$r = a \cdot q + b \cdot \ell = s \cdot u, \quad s = \sqrt{a^2 + b^2},\ u \sim \mathcal{N}(0,1)$$

> r of one base answer, in standard units u

$$\mathbb{E}[q \mid u] = \tfrac{a}{s} \cdot u, \quad \mathbb{E}[\ell \mid u] = \tfrac{b}{s} \cdot u, \quad \mathrm{Var}(\ell \mid u) = 1 - \tfrac{b^2}{s^2}$$

> q and ℓ given their reward (jointly normal)

$$\mathbb{E}[g] = \tfrac{a+b}{s} \cdot \mathbb{E}[M] - \tfrac{b^2}{s^2} \cdot \mathbb{E}[M^2] - \left(1 - \tfrac{b^2}{s^2}\right)$$

> M = the largest of n standard normals; $\mathbb{E}[M]$ and $\mathbb{E}[M^2]$ by one integral

Distance: the usual value for best-of-n is Kullback–Leibler (KL) = log n − (n − 1)/n; the plots use its square root.

Backup: the error bars

An average of m ratings is off by about its standard error.

$$\mathrm{SE} = \frac{\mathrm{sd}}{\sqrt{m}}$$

> **sd: the spread of people's score over kept answers**

$$n = 32: \ \mathrm{sd} = 2.08, \quad 2 \cdot \mathrm{SE} = 0.93 \ (m = 20), \quad 0.19 \ (m = 500)$$

> **the whole curve spans only 1.6**

Read: to halve the error bar, rate four times as many answers.